William Hewlett

One half of the duo who founded HP from their garage in Palo Alto in 1939, William Helwett was an electrical engineer from Stanford university. Their first successful product was a

precision audio oscillator and it just went on expanding from there. That expansion, atleast one division – and a major one at that – came to a halt this year. HP anonunced that it would discontinue its tablet business and sell off its PC division – despite being the #1 in 2010.

Voltage Modding

Voltage Modding is taking mere overclocking to a whole new level, by tweaking the voltages of components. Increasing voltages by a miniscule amount and stress testing the system, allows you to make your processor or graphics card run at a faster speed than its reference speed. A side effect of voltage modding, is generation of excess heat as when you make processors run faster, there are more transistors working faster, which generate more heat. Modding Graphics card is whole different ballgame, involving stuff such as soldering a resistor onto the PCB of the card. Needless to say, if you indulge in voltage modding, you automatically void the warranty on the card.

X79 Boards

Intel just launched its high end SAndy Bridge E series CPU in mid-November which will replace the older generation Intel Core i7-9xx extreme edition processors. It also brings a change of chipset from X58 to X79. Be prepared for the onslaught of X79 chipset based boards in the market, and of course a lot of reviews in Digit regarding the same. X79 boards house

the LGA2011 socket and have four dual channel DIMM slots, that is eight RAM slots. It has support for upto 2 x16 PCIe configurations which will please the gaming community.

SandForce Controller

Solid state drives are slowly but surely gaining momentum thank to prices coming down. Their performance though faster than a mechanical HDD, depends a lot on the type of controller they house, as this is the heard of an SSD. SandForce controller is a name to reckon with in this field. Infact our Zero1 Award went to an SSD sporting SandForce SF2281 controller. Some of the top SSDs in the market sport the SandForce controller.

Jumper

Anyone who has handled motherboard will be familiar with those tiny little plastic sheaths that cover some protruding pins on the board, which help in closing the circuit

for current flow. These are known as Jumpers and they are a very important accessory on the board. The most common jumper most of us use is the clear CMOS jumper, which tends to reset the BIOS to previous stable state. You also have jumpers to adjust voltages or adjust speed of memory, etc based on the motherboard.

1,400,000,000

That is the number of transistors that is expected to be housed in a space of 22 nano meter by Intel in its upcoming Ivy Bridge line of processors next year. To give you a perspective, the width of the human hair is 100,000 nanometers

LN2 Cooling

Overclocking is a way in which you can tend to heat up your processor to very high temperatures. While the mere mortals stay content with stock coolers, the slightly advanced user go for water based cooling mechanism, using radiators to drive away the heat from the CPU. But then comes a class of enthusiasts, who are professional overclockers, who go even beyond water cooling. Liquid Nitrogen (LN2) has a boiling point (yes, boiling, not freezing) of -195 degrees centigrade which is way beyond the freezing point of water. This quality of LN2 makes it ideal for using as a coolant in overclocking sessions, as it allows you to overclock to very high levels as temperature the

CPU falls below -100 degrees when it comes in contact with LN2. Recently the AMD 8-core processor FX-8150 was stretched from its base clock of 3.6GHz to over 8 GHz by overclockers using LN2 cooling.

Bulldozer

AMD came out with its FX series of processors with the flagship FX-8150 having eight physical cores. The codename for this line-up of high end desktop processors is Bulldozer. It is built on the 32-nm process. The basic building block called the Bulldozer module comprises two integer cores and a shared floating point core. Four such Bulldozer modules make up the eight cores. You can read more about the architecture in our November 2011 issue and read a review on the FX-8150 in the Bazaar section for this month.